

```

4
5 min_dist = float('inf')
6 closest_pair = None
7
8 for i in range(len(points)):
9     for j in range(i + 1, len(points)):
10         x1, y1 = points[i]
11         x2, y2 = points[j]
12
13         dist = math.sqrt((x2 - x1) ** 2 + (y2 - y1) ** 2)
14
15         if dist < min_dist:
16             min_dist = dist
17             closest_pair = (points[i], points[j])
18
19 print("Closest pair:", closest_pair[0], "-", closest_pair[1])
20 print("Minimum distance:", min_dist)

```

Closest pair: (1, 2) - (3, 1)
Minimum distance: 2.23606797749979

=== Code Execution Successful ===

```

4
5 min_dist = float('inf')
6 closest_pair = None
7
8 for i in range(len(points)):
9     for j in range(i + 1, len(points)):
10         x1, y1 = points[i]
11         x2, y2 = points[j]
12
13         dist = math.sqrt((x2 - x1) ** 2 + (y2 - y1) ** 2)
14
15         if dist < min_dist:
16             min_dist = dist
17             closest_pair = (points[i], points[j])
18
19 print("Closest pair:", closest_pair[0], "-", closest_pair[1])
20 print("Minimum distance:", min_dist)

```

Closest pair: (5, 4) - (4, 6)
Minimum distance: 2.23606797749979

=== Code Execution Successful ===

```

4
5 min_dist = float('inf')
6 closest_pair = None
7
8 for i in range(len(points)):
9     for j in range(i + 1, len(points)):
10         x1, y1 = points[i]
11         x2, y2 = points[j]
12
13         dist = math.sqrt((x2 - x1) ** 2 + (y2 - y1) ** 2)
14
15         if dist < min_dist:
16             min_dist = dist
17             closest_pair = (points[i], points[j])
18
19 print("Closest pair:", closest_pair[0], "-", closest_pair[1])
20 print("Minimum distance:", min_dist)

```

Closest pair: (2, 2) - (3, 1)
Minimum distance: 1.4142135623730951

=== Code Execution Successful ===

```
4
5 min_dist = float('inf')
6 closest_pair = None
7
8 for i in range(len(points)):
9     for j in range(i + 1, len(points)):
10         x1, y1 = points[i]
11         x2, y2 = points[j]
12
13         dist = math.sqrt((x2 - x1) ** 2 + (y2 - y1) ** 2)
14
15         if dist < min_dist:
16             min_dist = dist
17             closest_pair = (points[i], points[j])
18
19 print("Closest pair:", closest_pair[0], "-", closest_pair[1])
20 print("Minimum distance:", min_dist)
```

Closest pair: (10, 20) - (12, 22)
Minimum distance: 2.8284271247461903

=== Code Execution Successful ===

```

11
12     for i in range(len(points)):
13         for j in range(i + 1, len(points)):
14             d = distance(points[i], points[j])
15
16             if d < min_dist:
17                 min_dist = d
18                 pair = (points[i], points[j])
19
20     return pair, min_dist
21
22 points = [(1, 2), (4, 5), (7, 8), (3, 1)]
23
24 pair, dist = closest_pair(points)
25
26 print("Closest pair:", pair[0], "-", pair[1])
27 print("Minimum distance:", dist)

```

```

Closest pair: (1, 2) - (3, 1)
Minimum distance: 2.23606797749979

=== Code Execution Successful ===

```